WineWise – Wine Quality Predictor

DAT158 Final Project Report

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Project status: The WineWise application has been implemented, trained, evaluated, tested and publicly deployed using Streamlit Community Cloud.

Live application: https://winewise-dat158-aq76uxvru6mqgpaonzympd.streamlit.app

GitHub repository: https://github.com/678113/WineWise-DAT158

1: DESCRIBE THE PROBLEM

Scope

WineWise is a machine-learning-based web application that estimates the quality score of a red wine from eleven physicochemical measurements. Users enter values such as fixed and volatile acidity, residual sugar, sulphates, pH, density and alcohol concentration; the application returns a numerical prediction. The purpose is educational: to demonstrate an end-to-end supervised-learning workflow and to let users explore how a model responds to different wine characteristics.

The task is regression because the system predicts a numerical score. Machine learning is appropriate because the relationship between chemical properties and sensory quality may be complex and cannot easily be captured by a single manually defined rule. A non-ML alternative would be expert tasting, or using the average historical quality score as a crude estimate. The latter is represented by our Dummy Regressor baseline.

The intended users are students and curious users exploring wine data. The product provides an accessible interface to a trained model, not a replacement for professional sensory evaluation. Its components are the UCI dataset, data-loading and preprocessing code, model training and evaluation, a saved scikit-learn pipeline, and a Streamlit interface. The implementation runs locally using Python without specialized hardware.

Metrics

The primary model-selection metric is mean absolute error (MAE), measured using three-fold cross-validation on the training partition. MAE expresses the average absolute error in quality-score points and is directly understandable to users. Root mean squared error (RMSE) penalizes larger errors more strongly. R^2 measures predictive performance relative to predicting the test-set mean. Lower MAE and RMSE and higher R^2 are preferred.

Software success means the application starts, accepts valid inputs and returns a prediction from the saved model without errors. Project success is defined operationally as outperforming the mean-prediction baseline on cross-validation, producing reproducible test metrics, and providing a working interface. No user-satisfaction or commercial-impact study was conducted.

2: DATA

The project uses the Wine Quality dataset from the UCI Machine Learning Repository (Cortez et al., 2009). The red-wine subset was selected for the implemented model. The data consist of physicochemical measurements as numerical predictors and an observed quality rating as the supervised target.

The original red-wine dataset contained 1,599 observations. After removing duplicate rows, 1,359 unique observations remained for model development. The retained dataset was split into 1,087 training observations and 272 test observations using a fixed random seed of 42.

Predictors are fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. The target is quality. The CSV files were downloaded from UCI and placed in the local data directory. Labels come from the dataset rather than from users of the application; their accuracy and consistency are limited by the original sensory assessment process.

The scikit-learn preprocessing pipeline imputes missing numeric values using the training-set median and applies StandardScaler. Fitting preprocessing within each training fold avoids leakage of validation information. These preprocessing steps are included in the saved pipeline and applied consistently during training and prediction.

The data do not contain direct personal identifiers. Important limitations are that the model was trained only on red wine from the dataset's source region and production context, and that subjective quality ratings can vary between assessors. Predictions should not be treated as definitive judgments or as evidence of causal relationships between chemical properties and quality.

3: MODELING

Three regression approaches were evaluated: Dummy Regressor (predicting the training mean), Linear Regression and Random Forest Regressor. The Random Forest candidate used 120 trees, minimum leaf size 2, random state 42 and one worker. All models were wrapped in the same numeric preprocessing pipeline. The 20% test partition was held out during model selection.

The following results were obtained from three-fold cross-validation on the training data. The model with the lowest mean CV MAE was selected.

Model	CV MAE ↓	CV RMSE ↓
Dummy (mean)	0.6923	0.8190
Linear Regression	0.5145	0.6633
Random Forest	0.5099	0.6626

Random Forest had the lowest cross-validation MAE (0.5099), although its advantage over Linear Regression (0.5145) was small. The selected pipeline was refitted on all 1,087 training observations and evaluated once on the held-out 272 observations.

Final selected model	Test MAE ↓	Test RMSE ↓	Test R^2 ↑
Random Forest	0.4676	0.6161	0.4641

A test MAE of approximately 0.47 means that, on average, the predicted quality score differs from the recorded score by about 0.47 points in the held-out sample. An R^2 of 0.464 indicates that the model

captures some, but not all, of the variation in recorded quality. These results support a useful demonstration, not a claim of high-precision wine assessment.

A manual interface check kept the other displayed measurements unchanged and increased the alcohol input from 9.4 to 12.0. The prediction changed from 5.02 to 5.95. This verifies that the application responds to altered input, but it is not a controlled causal experiment or a general conclusion that increasing alcohol improves quality. A systematic residual analysis, feature-importance study and hyperparameter search were not evidenced by the available results and are opportunities for future work.

4: DEPLOYMENT

WineWise is implemented with Streamlit. The user fills out numeric wine-measurement fields and selects “Predict wine quality”. The application loads the trained Random Forest preprocessing-and-regression pipeline from models/wine_quality_pipeline.joblib and displays a predicted quality score. The saved pipeline ensures inference uses the same fitted transformations as model training.

The local application was launched successfully with “python -m streamlit run app.py”, and the user observed a working prediction screen. Automated tests were executed using python -m pytest -v, and all executed tests passed successfully.. The training command is “python -m src.train --kind red”, which produces the model artifact and results/metrics.json.

WineWise has been publicly deployed using Streamlit Community Cloud. The application is accessible at https://winewise-dat158-aq76uxvru6mqgpaonzympd.streamlit.app. The deployed application was manually tested by entering wine measurements and confirming that the trained model returned predictions. The prediction functionality was also tested after changing the alcohol input. The public GitHub repository at https://github.com/678113/WineWise-DAT158 contains the source code, trained model, evaluation metrics, tests, dependencies and documentation. This enables users to access the application and review the implementation.

Future maintenance would include checking input validity, tracking prediction failures, documenting the dataset version and dependencies, and retraining if the target population or input distributions change. Further improvements could include error analysis across quality levels, additional model tuning, and evaluated uncertainty estimates.

5: REFERENCES

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. Decision Support Systems, 47(4), 547–553. https://doi.org/10.1016/j.dss.2009.05.016

UCI Machine Learning Repository. Wine Quality dataset. https://archive.ics.uci.edu/dataset/186/wine+quality (accessed November 2026).

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AI assistance disclosure: ChatGPT was used to assist with project setup guidance, interpretation of experimental results, and drafting this report. The group is responsible for reviewing the code, verifying all statements against the repository, and ensuring that the submitted report accurately describes its own work.